

1.

如果 w 在 $\text{span}\{v_1, v_2\}$, 則必須存在
純量 c_1, c_2 , 使得 $c_1 v_1 + c_2 v_2 = w$

$$\begin{bmatrix} 2 & -4 & 8 \\ 3 & -5 & 2 \\ -5 & 8 & -9 \end{bmatrix} x = \begin{bmatrix} 6 \\ -7 \\ 11 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 4 \\ 3 & -5 & 2 \\ -5 & 8 & -9 \end{bmatrix} \begin{matrix} \\ \downarrow x(-3) \\ \downarrow x5 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -10 \\ 0 & -2 & 11 \end{bmatrix} \rightarrow \text{no solution}$$

Ans: w is not in the subspace

2. a. 3 個

b. ∞ 個

c.

$$\begin{matrix} x4 \downarrow \\ x(-3) \downarrow \end{matrix} \begin{bmatrix} 2 & -3 & -4 & 6 \\ -8 & 8 & 6 & -10 \\ 6 & -7 & -7 & 11 \end{bmatrix}$$

$$\sim \begin{matrix} \\ x2 \uparrow \end{matrix} \begin{bmatrix} 2 & -3 & -4 & 6 \\ 0 & -4 & -10 & 14 \\ 0 & 2 & 5 & -7 \end{bmatrix}$$

$$\sim \begin{matrix} \\ \uparrow \end{matrix} \begin{bmatrix} 2 & -3 & -4 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 5 & -7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -3 & -4 & 6 \\ 0 & 2 & 5 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} 2x_1 - 3x_2 - 4x_3 = 6 \\ 2x_2 + 5x_3 = -7 \end{cases}$$

$$\begin{cases} x_1 = \frac{1}{2}(-\frac{9}{2} - \frac{7}{2}x_3) \\ x_2 = \frac{1}{2}(-7 - 5x_3) \\ x_3 \text{ is free} \end{cases}$$

Ans: p is in Col A

3.

$$A = \begin{bmatrix} 3 & 2 & 1 & -5 \\ -9 & -4 & 1 & 7 \\ 9 & 2 & -5 & 1 \end{bmatrix}_{3 \times 4}$$

Ans: $p=4, q=3$

4.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ -5 & -1 & 0 \\ 2 & 7 & 11 \end{bmatrix}_{4 \times 3}$$

Ans: $p=3, q=4$

5.

for Col A

 $\therefore$ pivot 出现在 row 1, row 2

$$\therefore \left\{ \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} \right\}$$

for Null A

$$\begin{bmatrix} 1 & 2 & 6 & -5 & 0 \\ 0 & 1 & 5 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + 2x_2 + 6x_3 - 5x_4 = 0 \\ x_2 + 5x_3 - 6x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = 4x_3 - 7x_4 \\ x_2 = -5x_3 + 6x_4 \\ x_3 \text{ is free} \\ x_4 \text{ is free} \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4x_3 - 7x_4 \\ -5x_3 + 6x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Ans: a basis for Col A} = \left\{ \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} \right\}$$

$$\text{a basis for Null A} = \left\{ \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix} \right\}$$

b. for col A

$$\left\{ \begin{bmatrix} -3 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix} \right\}$$

for Nul A

$$\begin{bmatrix} 1 & -3 & 6 & 9 & 0 \\ 0 & 0 & 4 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 - 3x_2 + 6x_3 + 9x_4 = 0 \\ 4x_3 + 5x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = 3x_2 - \frac{3}{2}x_4 \\ x_2 \text{ is free} \\ x_3 = -\frac{5}{4}x_4 \\ x_4 \text{ is free} \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3x_2 - \frac{3}{2}x_4 \\ x_2 \\ -\frac{5}{4}x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{3}{2} \\ 0 \\ -\frac{5}{4} \\ 1 \end{bmatrix}$$

$$\text{Ans: a basis for Col A} = \left\{ \begin{bmatrix} -3 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix} \right\}$$

$$\text{a basis for Nul A} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{3}{2} \\ 0 \\ -\frac{5}{4} \\ 1 \end{bmatrix} \right\}$$

Compute the determinants in Exercises 7-8 using a cofactor expansion across the first row. In Exercises 7-8, also compute the determinant by a cofactor expansion down the second column.

$$7. \begin{vmatrix} 2 & -2 & 3 \\ 3 & 1 & 2 \\ 1 & 3 & -1 \end{vmatrix}$$

$$8. \begin{vmatrix} 1 & 2 & 4 \\ 3 & 1 & 1 \\ 2 & 4 & 2 \end{vmatrix}$$

7. ($\rightarrow$, col 1)

$$2 \cdot \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + 3 \cdot \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}$$

$$= -14 + 2 \cdot (-5) + 3 \cdot 8$$

$$= -14 - 10 + 24 = 0_{\#}$$

($\downarrow$, row 1)

$$2 \cdot \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} - 3 \cdot \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} + 1 \cdot \begin{vmatrix} -2 & 3 \\ 1 & 2 \end{vmatrix}$$

$$= -14 - 3 \cdot (-7) + (-7)$$

$$= -14 + 21 - 7 = 0_{\#}$$

($\rightarrow$, col 2)

$$(-3) \cdot \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + (-2) \cdot \begin{vmatrix} 2 & -2 \\ 1 & 3 \end{vmatrix}$$

$$= (-3) \cdot (-7) + (-5) + (-2) \cdot 8$$

$$= 21 - 5 - 16 = 0_{\#}$$

($\downarrow$, row 2)

$$2 \cdot \begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} - 3 \cdot \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix}$$

$$= 2 \cdot (-5) + (-5) - 3 \cdot (-5)$$

$$= -10 - 5 + 15 = 0_{\#}$$

8.

$(\rightarrow, \text{col } 1)$

$$1 \cdot \begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} - 2 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} + 4 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix}$$

$$= (-2) - 2 \cdot 4 + 4 \cdot 10$$

$$= -2 - 8 + 40 = 30 \quad \#$$

$(\downarrow, \text{row } 1)$

$$1 \cdot \begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} - 3 \cdot \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} + 2 \cdot \begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix}$$

$$= (-2) - 3 \cdot (-12) + 2 \cdot (-2)$$

$$= 30 \quad \#$$

$(\rightarrow, \text{col } 2)$

$$(-3) \cdot \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 4 \\ 2 & 2 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$= (-3) \cdot (-12) + (-6) - 0$$

$$= 36 - 6 = 30 \quad \#$$

$(\downarrow, \text{row } 2)$

$$(-2) \cdot \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 4 \\ 2 & 2 \end{vmatrix} + (-4) \cdot \begin{vmatrix} 1 & 4 \\ 3 & 1 \end{vmatrix}$$

$$= (-2) \cdot 4 + (-6) + (-4) \cdot (-11)$$

$$= -8 - 6 + 44 = 30 \quad \#$$

9.

$$\begin{vmatrix} 7 & 6 & 8 & 4 \\ 0 & 0 & 0 & 6 \\ 8 & 7 & 9 & 3 \\ 0 & 4 & 0 & 5 \end{vmatrix}$$

$$= 6 \cdot \begin{vmatrix} 7 & 6 & 8 \\ 8 & 7 & 9 \\ 0 & 4 & 0 \end{vmatrix}$$

$$= 6 \cdot (-4) \begin{vmatrix} 7 & 8 \\ 8 & 9 \end{vmatrix}$$

$$= (-24) \cdot (63 - 64)$$

$$= 24 \#$$

10.

$$\begin{vmatrix} 1 & -2 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 2 & -4 & -3 & 5 \\ 2 & 0 & 3 & 5 \end{vmatrix}$$

$$= (-3) \cdot \begin{vmatrix} 1 & -2 & 2 \\ 2 & -4 & 5 \\ 2 & 0 & 5 \end{vmatrix}$$

$$= (-3) \left[2 \cdot \begin{vmatrix} 2 & 5 \\ 2 & 5 \end{vmatrix} + (-4) \cdot \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} \right]$$

$$= (-3) \cdot (-4) = 12 \#$$

Linear Algebra

線性代數

Homework_5

1. Let $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -4 \\ -5 \\ 8 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 8 \\ 2 \\ -9 \end{bmatrix}$. Determine if $\mathbf{w}$ is in the subspace of $\mathbb{R}^3$ generated by $\mathbf{v}_1$ and $\mathbf{v}_2$.

2. Let $\mathbf{v}_1 = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -3 \\ 8 \\ -7 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} -4 \\ 6 \\ -7 \end{bmatrix}$,

$$\mathbf{p} = \begin{bmatrix} 6 \\ -10 \\ 11 \end{bmatrix}, \text{ and } A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3].$$

- How many vectors are in $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?
- How many vectors are in $\text{Col } A$?
- Is $\mathbf{p}$ in $\text{Col } A$? Why or why not?

In Exercises 3 and 4, give integers p and q such that $\text{Nul } A$ is a subspace of $\mathbb{R}^p$ and $\text{Col } A$ is a subspace of $\mathbb{R}^q$.

$$3. A = \begin{bmatrix} 3 & 2 & 1 & -5 \\ -9 & -4 & 1 & 7 \\ 9 & 2 & -5 & 1 \end{bmatrix}$$

$$4. A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ -5 & -1 & 0 \\ 2 & 7 & 11 \end{bmatrix}$$

Exercises 5-6 display a matrix A and an echelon form of A . Find a basis for $\text{Col } A$ and a basis for $\text{Nul } A$.

$$5. A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$6. A = \begin{bmatrix} -3 & 9 & -2 & -7 \\ 2 & -6 & 4 & 8 \\ 3 & -9 & -2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 6 & 9 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Compute the determinants in Exercises 7-8 using a cofactor expansion across the first row. In Exercises 7-8, also compute the determinant by a cofactor expansion down the second column.

$$7. \begin{vmatrix} 2 & -2 & 3 \\ 3 & 1 & 2 \\ 1 & 3 & -1 \end{vmatrix}$$

$$8. \begin{vmatrix} 1 & 2 & 4 \\ 3 & 1 & 1 \\ 2 & 4 & 2 \end{vmatrix}$$

Compute the determinants in Exercises 9-10 by cofactor expansions. At each step, choose a row or column that involves the least amount of computation.

$$\mathbf{9.} \quad \begin{vmatrix} 7 & 6 & 8 & 4 \\ 0 & 0 & 0 & 6 \\ 8 & 7 & 9 & 3 \\ 0 & 4 & 0 & 5 \end{vmatrix} \quad \mathbf{10.} \quad \begin{vmatrix} 1 & -2 & 4 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -4 & -3 & 5 \\ 2 & 0 & 3 & 5 \end{vmatrix}$$